Econometrics Chapter 5: OLS using Matrix Algebra

(1) Setting up the Model

Population Setup

Population: $E(Y|X_1, X_2, \ldots, X_{k-1}) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_{k-1} X_{k-1}$

ε: $Y - \beta_0 - \beta_1 X_1 - \beta_2 X_2 - \ldots - \beta_{k-1} X_{k-1}$

- True Population Model:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_{k-1} X_{k-1} + \varepsilon$$

$E(\varepsilon|X_1, X_2, \ldots, X_{k-1}) = 0$

whereby, $X_1 \ldots X_{k-1}$ refers to K-1 different regressors

Sample Setup

- representative iid sample from the population

$\{Y_i, X_{i1}, X_{i2}, \ldots, X_{i,k-1}\}_{i=1}^{n}$

- Sample Model:

$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \ldots + \beta_{k-1} X_{i,k-1} + \varepsilon_i$ for all $i = 1, \ldots, n$

- Errors:

$E(\varepsilon_i | X_{11} \ldots X_{n1}, X_{12} \ldots X_{n2}, \ldots, X_{1k-1}, \ldots X_{nk-1}) = 0$

$E(\varepsilon_i^2 | X_{11} \ldots X_{n1}, X_{12} \ldots X_{n2}, \ldots, X_{1k-1}, \ldots X_{nk-1}) = \sigma_i^2$ or σ^2

$E(\varepsilon_i \varepsilon_j | X_{11} \ldots X_{n1}, X_{12} \ldots X_{n2}, \ldots, X_{1k-1}, \ldots X_{nk-1}) = 0$

for all $i, j = 1, \ldots, n$, $i \neq j$

- Linear Algebraic Expression of the Model

$$\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} = \begin{bmatrix} 1 & X_{11} & X_{12} & \ldots & X_{1k-1} \\ 1 & X_{21} & X_{22} & \ldots & X_{2k-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & X_{n2} & \ldots & X_{n,k-1} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_{k-1} \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_{k-1} \end{bmatrix}$$

$$y = X\beta + \varepsilon$$

whereby, X_{ij} is the ith observation of the jth regressor

Partitioning the Regressor Matrix X

- Partition by Column

$$X = \begin{bmatrix} 1 & X_{11} & X_{12} & \cdots & X_{1,K-1} \\ 1 & X_{21} & X_{22} & \cdots & X_{2,K-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & X_{n2} & \cdots & X_{n,K-1} \end{bmatrix} = \begin{bmatrix} X_{1*} \\ X_{2*} \\ \vdots \\ X_{n*} \end{bmatrix}$$

whereby,

$$\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} = \begin{bmatrix} X_{1*} \\ X_{2*} \\ \vdots \\ X_{n*} \end{bmatrix} \beta + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_{K-1} \end{bmatrix} \quad \text{or} \quad Y_i = X_{i*}\beta + \varepsilon_i \;, i = 1, 2, \ldots, n$$

- Partition by Row (Alternate Partitioning)

$$X = \left[\begin{array}{c|c|c|c|c} 1 & X_{11} & X_{12} & \cdots & X_{1,K-1} \\ 1 & X_{21} & X_{22} & \cdots & X_{2,K-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & X_{n2} & \cdots & X_{n,K-1} \end{array}\right] = \begin{bmatrix} i_n & X_{*1} & X_{*2} & \cdots & X_{*,K-1} \end{bmatrix}$$

Collinearity Condition

- whereby (partition by Row)

$$Xc = c_0 + c_1 X_{*1} + c_2 X_{*2} + \ldots + c_{K-1} X_{*,K-1} = 0_{n\times 1} \text{ iff } c = 0_K$$

where $c = [c_0 \; c_1 \; c_2 \ldots c_{K-1}]$

- ols estimation requires no perfect collinearity

- Assumption can also be written as:

$Xc \neq 0_n \Leftrightarrow c \neq 0_K$

Linear Algebraic Form of Error

- $E(\varepsilon_i | X_{11} \dots X_{n1}, X_{12} \dots X_{n2}, \dots, X_{1K-1} \dots X_{nK-1}) = 0$

can be written as

$$\begin{bmatrix} E(\varepsilon_1|X) \\ E(\varepsilon_2|X) \\ \vdots \\ E(\varepsilon_n|X) \end{bmatrix} = 0_n \qquad E(\varepsilon_i|X) = 0$$

- $E(\varepsilon_i^2 | X_{11} \dots X_{n1}, X_{12} \dots X_{n2}, \dots, X_{1K-1}, \dots X_{nK-1}) = \sigma_i^2$ or σ^2

$E(\varepsilon_i \varepsilon_j | X_{11} \dots X_{n1}, X_{12} \dots X_{n2}, \dots, X_{1K-1}, \dots X_{nK-1}) = 0$

can be written as

(Var-Cov Matrix of Error)

$$\begin{bmatrix} E(\varepsilon_1^2|X) & E(\varepsilon_1\varepsilon_2|X) & \dots & E(\varepsilon_1\varepsilon_n|X) \\ E(\varepsilon_1\varepsilon_2|X) & E(\varepsilon_2^2|X) & \dots & E(\varepsilon_2\varepsilon_n|X) \\ \vdots & \vdots & \ddots & \vdots \\ E(\varepsilon_1\varepsilon_n|X) & E(\varepsilon_2\varepsilon_n|X) & \dots & E(\varepsilon_n^2|X) \end{bmatrix} = \begin{bmatrix} \sigma_1^2 & 0 & \dots & 0 \\ 0 & \sigma_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_n^2 \end{bmatrix}$$

$$Var(\varepsilon|X) = E(\varepsilon\varepsilon^T|X) = diag(\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2)$$

- If noise terms are homoskedastic

[variance of error is constant across all level of independent variable]

$$E(\varepsilon\varepsilon^T|X) = \begin{bmatrix} \sigma^2 & 0 & \dots & 0 \\ 0 & \sigma^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma^2 \end{bmatrix}$$

(2) OLS Estimation of MLR using Linear Algebra (Matrix Form)

MLR Model: $Y = X\beta + \varepsilon$

$$E(\varepsilon|X) = 0$$

$$E(\varepsilon\varepsilon^T|X) = \sigma^2 I_n$$

For any potential estimator $\hat{\beta}$,

- Fitted values: $\hat{y} = X\hat{\beta}$
- Residuals: $\hat{\varepsilon} = y - \hat{y} = y - X\hat{\beta}$
- Residual Sum of Squares $RSS(\hat{\beta})$: $\sum \hat{\varepsilon}_i^2 = \hat{\varepsilon}^T\hat{\varepsilon}$

$$\begin{aligned} -RSS(\hat{\beta}) &= \hat{\varepsilon}^T\hat{\varepsilon} \\ &= (y - X\hat{\beta})^T(y - X\hat{\beta}) \\ &= (y^T - \hat{\beta}^T X^T)(y - X\hat{\beta}) \\ &= y^Ty - y^TX\hat{\beta} - \hat{\beta}^TX^Ty - \hat{\beta}^TX^TX\hat{\beta} \\ &= y^Ty - 2\hat{\beta}X^Ty + \hat{\beta}^TX^TX\hat{\beta} \end{aligned}$$

OLS Argument:

$$-\underset{\hat{\beta}}{\operatorname{argmin}}\, RSS(\hat{\beta}) = \underset{\hat{\beta}}{\operatorname{argmin}}\; y^Ty - 2\hat{\beta}X^Ty + \hat{\beta}^TX^TX\hat{\beta}$$

- FOC: $\dfrac{\partial SSR(\hat{\beta})}{\partial \beta} = -2X^Ty + 2X^TX\hat{\beta}$

SOC: $\dfrac{\partial^2 SSR(\hat{\beta})}{\partial \beta \partial \beta^T} = 2X^TX$

- Solving: $-2X^Ty + 2X^TX\hat{\beta}^{ols} = 0_k$

$$\hat{\beta}^{ols} = (X^TX)^{-1}X^Ty$$

- $Xc \neq 0$ and $c \neq 0$ meant that X^TX is non-singular
- $c^TX^TXc = (Xc)^T(Xc) > 0$ (Positive Definite)

so $\hat{\beta}^{ols}$ minimizes $SSR(\hat{\beta})$

Partitioning X into columns [Form that emphasizes observations]

- Allow $X = \begin{bmatrix} X_{1*} \\ X_{2*} \\ \vdots \\ X_{n*} \end{bmatrix}$

- Therefore,

$$\hat{\beta}^{ols} = (X^TX)^{-1}X^T$$

$$= \left\{ \begin{bmatrix} X_{1*} & X_{2*} & \dots & X_{n*} \end{bmatrix} \begin{bmatrix} X_{1*} \\ X_{2*} \\ \vdots \\ X_{n*} \end{bmatrix} \right\}^{-1} \begin{bmatrix} X_{1*} & X_{2*} & \dots & X_{n*} \end{bmatrix} \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}$$

$$= \underbrace{\left(\sum_{i=1}^{n} X_{i*}^T X_{i*} \right)^{-1}}_{k \times k} \underbrace{\left(\sum_{i=1}^{n} X_{i*}^T Y_i \right)}_{k \times 1}$$

OLS Fitted Values

$$\hat{y}^{ols} = X\hat{\beta}^{ols}$$

$$= X(X^TX)^{-1}X^Ty$$

$$= Py \quad \text{whereby,} \quad P = (X(X^TX)^{-1}X^T)$$

OLS Residuals

$$\hat{\varepsilon}^{ols} = y - \hat{y}^{ols}$$

$$= y - X\hat{\beta}^{ols}$$

$$= y - X(X^TX)^{-1}X^Ty$$

$$= (I_n - X(X^TX)^{-1}X^T)y$$

$$= My$$

$$M = I_n - X(X^TX)^{-1}X^T$$

- simply FOC: $X^Ty - X^TX\hat{\beta}^{ols} = 0$

$$X^T(y - X\hat{\beta}^{ols}) = 0$$

$$X^T\hat{\varepsilon}_{ols} = 0$$

Error in Matrix Form

- $X = [i_n \; X_{*1} \; X_{*2} \; \dots \; X_{*K-1}]$ (partioned in rows)

$$X^T\hat{\varepsilon} = [i_n \; X_{*1} \; X_{*2} \; \dots \; X_{*K-1}] \begin{bmatrix} \hat{\varepsilon}_i^{ols} \\ \hat{\varepsilon}_i^{ols} \\ \vdots \\ \hat{\varepsilon}_i^{ols} \end{bmatrix}$$

$$= \begin{bmatrix} \sum \hat{\varepsilon}_i^{ols} \\ \sum X_{i1}\hat{\varepsilon}_i^{ols} \\ \vdots \\ \sum X_{iK-1}\hat{\varepsilon}_i^{ols} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

- Implications: $\sum \hat{\varepsilon}_i^{ols} = 0$ [First line]

$$\frac{1}{n}\hat{\varepsilon}_i^{ols} = \bar{\hat{\varepsilon}}_i^{ols} = 0$$

$$\sum_{i=1}^{n} X_{ij}\hat{\varepsilon}_i^{ols} = 0 \quad \text{sample } cov(X_{ij}, \hat{\varepsilon}_i^{ols}) = 0$$

for each X_j, $j = 1, \dots, K-1$

(3) Goodness of Fit Test

$TSS = FSS + RSS$ [Holds in general MLR]

Proof:

- Allow $M_0 = I_n - (\frac{1}{n}) i_n i_n^T$ which is symmetric and idempotent
  - $M_0^T = M_0$
  - $M_0 M_0 = M_0$
- Important Results:
  - i. $M_0 y = \begin{bmatrix} Y_1 - \bar{Y} \\ Y_2 - \bar{Y} \\ \vdots \\ Y_n - \bar{Y} \end{bmatrix}$
  - ii. $y^T M_0^T M_0 y = y^T M_0 y = \sum_{i=1} (Y_i - \bar{Y})^2$
  - iii $M_0 \hat{\varepsilon}^{ols} = \hat{\varepsilon}^{ols}$
- Deriving Goodness of fit

$$y = \hat{y}^{ols} + \hat{\varepsilon}^{ols}$$

$$M_0 y = M_0 \hat{y}^{ols} + M_0 \hat{\varepsilon}^{ols}$$

$$M_0 y = M_0 \hat{y}^{ols} + \hat{\varepsilon}^{ols}$$

$$y^T M_0^T M_0 y = (M_0 \hat{y}^{ols} + \hat{\varepsilon}^{ols})^T (M_0 \hat{y}^{ols} + \hat{\varepsilon}^{ols})$$

$$y^T M_0 y = \hat{y}_{ols}^T M_0 \hat{y}^{ols} + \hat{y}_{ols}^T M_0 \hat{\varepsilon}^{ols} + \hat{\varepsilon}^{ols T} M_0 \hat{y}^{ols} + \hat{\varepsilon}_{ols}^T \hat{\varepsilon}^{ols}$$

$$y^T M_0 y = \hat{y}_{ols}^T M_0 \hat{y}^{ols} + \hat{\varepsilon}_{ols}^T \hat{\varepsilon}_{ols}$$

$$TSS = FSS + RSS \quad [SST = SSE + SSR]$$

$\hat{y}^T \hat{\varepsilon} = (X\hat{\beta})(y - X\hat{\beta})$

$= (X^T \hat{\varepsilon})\hat{\beta}$

$= 0$

$$R^2 = 1 - \frac{RSS}{TSS}$$

$$\text{Adj } R^2 = 1 - \frac{\frac{1}{n-k} RSS}{\frac{1}{n-1} TSS} = 1 - \frac{RSS}{TSS} \frac{n-1}{n-k}$$

(4) Properties of OLS Estimators

1. OLS Estimator $\hat{\beta}_{ols}$ is unbiased

$$\hat{\beta}_0^{ols} = (X^TX)^{-1}X^Ty$$
$$= (X^TX)^{-1}X^T(X\beta + \varepsilon)$$
$$= (X^TX)^{-1}X^TX\beta + (X^TX)^{-1}X^T\varepsilon$$
$$= \beta + (X^TX)^{-1}X^T\varepsilon$$

$$E(\hat{\beta}_{ols}|X) = \beta + (X^TX)^{-1}X^TE(\varepsilon|X)$$
$$= \beta$$

$$E(\hat{\beta}_{ols}) = \beta$$

2. OLS Estimator Variance [Homoskedasticity case]

- given $Var(\varepsilon\varepsilon^T|X) = \sigma^2 I_n$

$$Var(\hat{\beta}^{ols}|X) = Var(\beta + (X^TX)^{-1}X^T\varepsilon|X)$$
$$= (X^TX)^{-1}X^TVar(\varepsilon|X)X(X^TX)^{-1}$$
$$= (X^TX)^{-1}X^T\sigma^2X(X^TX)^{-1}$$
$$= \sigma^2(X^TX)^{-1}$$

- require the estimate of σ^2

unbiased estimator of σ^2 is $\hat{\sigma}^2$

$$\hat{\sigma}^2 = \frac{1}{n-K}\sum_{i=1}^{n}\hat{\varepsilon}_i^{\,2}{}_{ols}$$
$$= \frac{\hat{\varepsilon}_{ols}^T\hat{\varepsilon}_{ols}}{n-k}$$

Proof & Derivation of unbiased estimator of σ^2

- Given $\hat{\varepsilon}_{ols} = My$ whereby $M = I_n - X(X^TX)^{-1}X^T$
  - $MX = 0$
  - M is symmetric
  - M is idempotent $MM = M$

$$\hat{\varepsilon}^{ols} = My$$
$$= M(X\beta + \varepsilon)$$
$$= M\varepsilon$$
$$\hat{\varepsilon}^T_{ols}\hat{\varepsilon}_{ols} = \varepsilon^T M^T M\varepsilon$$
$$= \varepsilon^T M\varepsilon$$
$$E(\hat{\varepsilon}^T_{ols}\hat{\varepsilon}_{ols}|X) = E(\varepsilon^T M\varepsilon|X)$$
$$= E(\text{trace}(\varepsilon^T M\varepsilon)|X)$$
$$= E(\text{trace}(\varepsilon\varepsilon^T M)|X)$$
$$= \text{Trace}\, E(\varepsilon\varepsilon^T M|X)$$
$$= \text{Trace}[E(\varepsilon\varepsilon^T|X)M]$$
$$= \text{trace}(\sigma^2 I_n M)$$
$$= \sigma^2 \text{trace}(M)$$
$$= \sigma^2 \text{trace}(I_n - X(X^TX)^{-1}X^T)$$
$$= \sigma^2[\text{trace}(I_n) - \text{trace}(X(X^TX)^{-1}X^T)$$
$$= \sigma^2(n - \text{trace}(I_k)$$
$$= \sigma^2(n-k)$$
$$E\left(\frac{\hat{\varepsilon}^T_{ols}\hat{\varepsilon}_{ols}}{n-k}\middle| X\right) = \sigma^2$$
$$E\left(\frac{\hat{\varepsilon}^T_{ols}\hat{\varepsilon}_{ols}}{n-k}\right) = \sigma^2$$

3. OLS Estimator Variance [Heteroskedasticity case]

- $Var(\hat{\beta}^{ols}|X) = (X^TX)^{-1}X^T Var(\varepsilon|X) X (X^TX)^{-1}$

Express X partition by row

$$Var(\hat{\beta}^{ols}|X) = \left(\sum X_{i*}^T X_{i*}\right)^{-1} \begin{bmatrix} X_{1*}^T & X_{2*}^T & \cdots & X_{n*}^T \end{bmatrix} \begin{bmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \\ 0 & 0 & \cdots & \sigma_n^2 \end{bmatrix} \begin{bmatrix} X_{1*} \\ X_{2*} \\ \vdots \\ X_{n*} \end{bmatrix} \left(\sum X_{i*}^T X_{i*}\right)^{-1}$$

$$= \left(\sum X_{i*}^T X_{i*}\right)^{-1} \left(\sum \sigma_i^2 X_{i*}^T X_{i*}\right) \left(\sum X_{i*}^T X_{i*}\right)^{-1}$$

- Heteroskedasticity-Robust Estimator Variance-Covariance Matrix

($\hat{\varepsilon}_i^2{}_{ols}$ to approximate σ_i^2)

$$\widehat{Var}_{HCO}(\hat{\beta}^{ols}) = \left(\sum X_{i*}^T X_{i*}\right)^{-1} \left(\sum \hat{\varepsilon}_{i,ols}^2 X_{i*}^T X_{i*}\right) \left(\sum X_{i*}^T X_{i*}\right)^{-1}$$

4. $\hat{\beta}^{ols}$ is BLUE under homoskedasticity

- unbiasness shown
- linearity:

$$\hat{\beta}ols = \underbrace{(X^TX)^{-1}X^T}_{A} y$$

- BLUE — smallest variance amongst all linear unbiased estimators
- let $\tilde{\beta} = By$ whereby $B \neq (X^TX)^{-1}X^T y$

$$B = D + (X^TX)^{-1}X^T$$

$$\tilde{\beta} = By$$

$$= (D + (X^TX)^{-1}X^T)y$$

$$= (D + (X^TX)^{-1}X^T)(X\beta + \varepsilon)$$

$$= DX\beta + D\varepsilon + (X^TX)^{-1}X^TX\beta + (X^TX)^{-1}X^T\varepsilon$$

- to ensure unbiasness of $\tilde{\beta}$, assume $DX = 0$

$$\begin{aligned}
\text{Var}(\tilde{\beta}|X) &= E((\tilde{\beta}-\beta)(\tilde{\beta}-\beta)^T|X) \\
&= E((D\varepsilon + (X^TX)^{-1}X^T\varepsilon)(D\varepsilon + (X^TX)^{-1}X^T\varepsilon)^T|X) \\
&= D + (X^TX)^{-1}X^T E(\varepsilon\varepsilon^T|X)(D + (X^TX)^{-1}X^T)^T \\
&= \sigma^2(D+(X^TX)^{-1}X^T)(D+(X^TX)^{-1}X^T)^T \\
&= \sigma^2(D + (X^TX)^{-1}X^T)(D^T + X(X^TX)^{-1}) \\
&= \sigma^2(DD^T + DX(X^TX)^{-1} + (X^TX)^{-1}X^TD + (X^TX)^{-1}) \\
&= \sigma^2(DD^T + (X^TX)^{-1} \\
&= \sigma^2 DD^T + \text{Var}(\hat{\beta}^{ols}|X)
\end{aligned}$$

Therefore: (Ensuring DD^T is +ve)

$$\begin{aligned}
\text{Var}(c^T\tilde{\beta}|X) &= c^T\text{Var}(\tilde{\beta}|X)c \\
&= c^T(\sigma^2 DD^T + \text{Var}(\hat{\beta}^{ols}|X))c \\
&= \sigma^2 c^T DD^T c + \text{Var}(c^T\hat{\beta}^{ols}|X) \\
&= \sigma^2(D^Tc)^T D^Tc + \text{Var}(c^T\hat{\beta}^{ols}|X) \geq \text{Var}(c^T\hat{\beta}^{ols}|X)
\end{aligned}$$

$\therefore D^Tc$ is a vector

therefore, $(D^Tc)^TD^Tc$ is a sum of squares which cannot be neg

(5) Asymptotic Properties of OLS Estimators

Multivariate versions of LLN and CLT

- Khinchine's LLN: If $\{z_i\}_{i=1}^n$ iid, $E(z_i)=\mu<\infty$, then $\bar{z} \xrightarrow{p} \mu$
- Lindeberg-Levy CLT: If $\{z_i\}_{i=1}^n$ iid, $E(z_i)=\mu$ and $Var(z_i)=\sigma^2<\infty$

  then $\sqrt{n}(\bar{z}-\mu) \xrightarrow{d} N(0,\sigma^2)$

  - If $\mu=0$, $\sqrt{n}\bar{z} = \frac{1}{\sqrt{n}}\sum z_i \xrightarrow{d} N(0,\sigma^2)$

- Extrapolating these LLN & CLT into multivariate versions

If $\{z_i\}_{i=1}^n$ iid vectors of random variables with $E(z_i)=0$ and $Var(z_i)=\Omega$ for all i,

$$\bar{z} = \frac{1}{n}\sum z_i = \begin{bmatrix} (1/n)\sum z_{1i} \\ (1/n)\sum z_{2i} \\ \vdots \\ (1/n)\sum z_{ki} \end{bmatrix} \xrightarrow{p} \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_k \end{bmatrix} = \mu$$

$$\sqrt{n}\bar{z} = \frac{1}{\sqrt{n}}\sum z_i = \begin{bmatrix} (1/\sqrt{n})\sum z_{1i} \\ (1/\sqrt{n})\sum z_{2i} \\ \vdots \\ (1/\sqrt{n})\sum z_{ki} \end{bmatrix} \xrightarrow{d} \text{Normal}_k(0,\Omega)$$

Limiting Distributions — how an estimator behave as $n\to\infty$

- sub z to be $\hat{\beta}$

$$\sqrt{n}(\hat{\beta}-\beta) = \left(\frac{1}{n}\sum X_{i*}^T X_{i*}\right)^{-1}\left(\frac{1}{\sqrt{n}}\sum X_{i*}^T \varepsilon_i\right)$$

given $\frac{1}{\sqrt{n}}\sum X_{i*}\varepsilon_i \xrightarrow{d} \text{Normal}_{k+1}(0,S)$

then $\sqrt{n}(\hat{\beta}^{ols}-\beta) = \left(\frac{1}{n}\sum X_{i*}^T X_{i*}\right)^{-1}\left(\frac{1}{\sqrt{n}}\sum X_{i*}^T \varepsilon_i\right)$

$$\xrightarrow{d} \text{Normal}_{k+1}(0, \Sigma_{xx}^{-1} S \Sigma_{xx}^{-1})$$

$$Var(\hat{\beta}^{ols}) \approx \left(\frac{1}{n}\sum x x^{-1} S \sum x x^{-1}\right) \downarrow$$

- estimator of $\hat{\Sigma}_{xx} = \frac{1}{n}\sum X_{i*}^T X_{i*} = \frac{1}{n}X^T X$ — pop covariance of X

estimator of $\hat{S} = \frac{1}{n}\sum \hat{\varepsilon}_i^2 X_{i*}^T X_{i*} \xrightarrow{p} S$

Estimating Asymptotic Variance of $\hat{\beta}$

- whereby $A = \frac{1}{n}\sum X_{i*}^T X_{i*} = \hat{\Sigma}_{xx}$ and $B = \hat{S}$

$$\widehat{AVar}(\hat{\beta}^{ols}) = ABA^{-1}$$

$$= \left(\frac{1}{n}\sum X_{i*}^T X_{i*}\right)^{-1}\left(\frac{1}{n}\sum \hat{\varepsilon}_i^2 X_{i*}^T X_{i*}\right)\left(\frac{1}{n}\sum X_{i*}^T X_{i*}\right)^{-1}$$

- justifies the use of $\widehat{Var}_{HCO}(\hat{\beta})$

$$\widehat{Var}_{HCO}(\hat{\beta}^{ols}) = \frac{1}{n}\widehat{AVar}(\hat{\beta}^{ols})$$

$$= \left(\sum X_{i*}^T X_{i*}\right)^{-1}\left(\sum \hat{\varepsilon}_i^2 X_{i*}^T X_{i*}\right)\left(\sum X_{i*}^T X_{i*}\right)^{-1}$$